Speaker-Preserving Speech-to-Singing Synthesis with a WORLD-Residual Conformer

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***Abstract*—We generate a person's singing from their speech alone: they read the lyrics, we pick a melody, and the system sings the lyrics to that melody in their voice. Frozen HuBERT, ECAPA-TDNN and WORLD features feed a small Transformer that predicts how long each sound is held and a Conformer acoustic model, followed by a pretrained NSF-HiFiGAN vocoder. In our hybrid mode the Conformer corrects a classical WORLD conversion instead of generating the spectrogram from scratch. On NUS-48E, with validation and test speakers never seen in training, the hybrid reaches a mel-cepstral distortion of 4.12 dB against 4.32 dB for WORLD, the best pitch accuracy of all systems (F0 correlation 0.93) and a speaker similarity of 0.35 against 0.44 for real singing, whereas a purely neural model loses the speaker. It follows other singers' melodies without picking up their voices and keeps the voices of two authors recorded outside the corpus. Our experiments point to a trade-off between spectral accuracy and speaker identity, and keeping the melody's rests in place makes fully automatic synthesis 0.5 dB better.**

***Index Terms*—speech-to-singing conversion, singing voice synthesis, speaker preservation, HuBERT, Conformer, WORLD**

I. Introduction

Systems such as DiffSinger [1] produce convincing singing, but only after training on hours of studio recordings of the target singer, while most people have only ever recorded their speech. In *speech-to-singing* synthesis the person reads the lyrics, a melody is chosen separately, and the system sings the lyrics to that melody in the person's voice.

Singing changes three things: vowels are held for the length of a note (a sung phone lasts 2.3 times as long as a spoken one in our data), pitch moves from free glides to stable notes, and the voice takes on a singing resonance while it should still sound like the same person. Signal-processing methods [2] keep the timbre but sound artificial. Neural methods [3] learn the mapping, but with little data they drift towards an average voice. We ran into exactly this: our first, purely neural model matched the spectrum well and lost the speaker entirely, which led to the design in this paper. Our main contributions are:

- a pipeline from raw audio that trains only a 393k-parameter duration predictor and a 9.5M-parameter Conformer on 40 songs, with HuBERT [4], ECAPA-TDNN [5] and NSF-HiFiGAN [6], [7] kept frozen;
- a *hybrid* model that corrects a WORLD [8] conversion, which brings back the speaker identity the neural model loses;
- experiments on own, transposed and other singers' melodies, unseen speakers, ablations, adversarial fine-tuning and regularisation, showing a trade-off between spectral accuracy and speaker identity;
- cross-singer alignment, rest-anchored timing, a post-filter and a forced-alignment front end for new recordings, tested on our own voices.

Related work. Saitou et al. [2] converted speech into singing with STRAIGHT analysis and separate F0, duration and spectrum models; Parekh et al. [3] learned the conversion with an encoder-decoder network. Both rely on parallel corpora such as NUS-48E [9] and NHSS [10]. DiffSinger [1] refines a rough mel spectrogram with a shallow diffusion step, close in spirit to our residual refinement, and the neural source-filter HiFi-GAN [6], [7] we use is driven by F0, which suits singing.

II. Dataset

We use NUS-48E [9]. Twelve speakers, six male and six female, each read and sang the lyrics of four English songs, which gives 48 paired recordings of 20 different songs (about 169 minutes) with phone-level annotations for both versions. We train on ten speakers (40 songs); JLEE (validation) and VKOW (test) are never seen in training.

Three observations shaped the system. Inside each pair the spoken and sung phone sequences line up one to one (only silence markers differ), which gives the exact spoken and sung length of every phone (Fig. 1, left). The pitch tracker mattered more than expected: CREPE-tiny [11] failed on about half of the singers and gave a note to only 54% of the sung vowels, so we switched to WORLD's Harvest estimator (99.8%). And

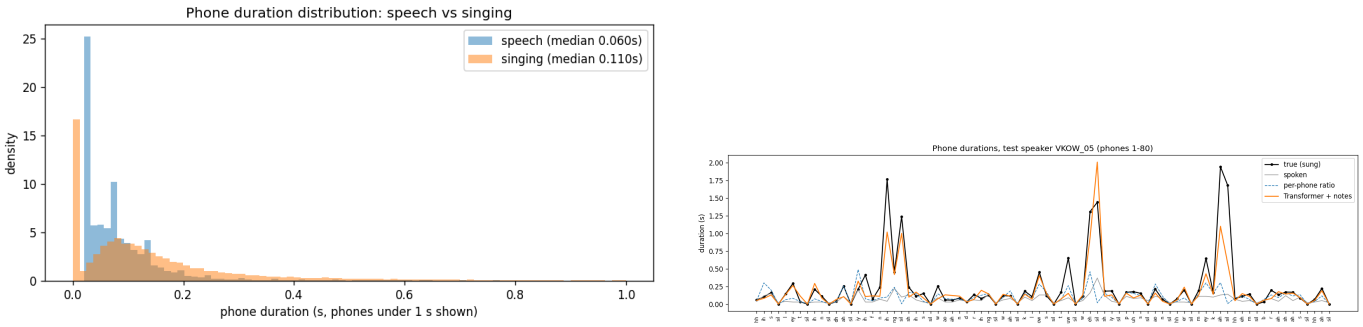


Fig. 1. Left: phone durations in NUS-48E, with a long tail of held sung notes. Right: true and predicted sung durations of 60 phones of a test song.

two singers' versions of the same song share only 77–89% of their phones (Section III-D).

III. Method

Fig. 2 gives an overview: the system takes the spoken lyrics and a sung reference of the same song, possibly by another person, each with its phone annotation.

A. Features and WORLD baseline

From the speech, resampled to 16 kHz, we take HuBERT-base layer-9 features (768-d, 50 frames per second), which carry phonetic content with little speaker information, and a 192-d ECAPA-TDNN speaker embedding. From the reference we take the Harvest F0 (60–1100 Hz, 100 frames per second, 0 Hz for unvoiced frames) and split it into notes: stable-pitch stretches of at least three frames, rounded to MIDI pitches. The model predicts the 128-bin log-mel spectrogram of the real singing, computed with the vocoder's own STFT (44.1 kHz, hop 512, 86.1 frames per second). Our WORLD baseline analyses the speech, stretches each phone's spectral envelope and aperiodicity to its sung length, swaps in the melody's F0 and resynthesises. It keeps the timbre but sounds buzzy, and the same conversion is the input of the hybrid model.

B. Duration predictor

For each spoken phone the predictor sees a learned 64-d phone embedding, the log spoken duration, the HuBERT features averaged over the phone, and the MIDI pitch, log length and a has-note flag of the note it falls on. Notes are matched to phones through the melody's annotation, the information a score with lyrics would give. A two-layer Transformer encoder ($d = 128$, 4 heads) predicts the log sung duration and is trained with

$$\mathcal{L}_{\text{dur}} = \tfrac{1}{N} \textstyle\sum_i \left(\log \hat{d}_i - \log d_i\right)^2. \tag{1}$$

C. Acoustic model and vocoder

Everything the acoustic model sees is placed on the singing timeline. A sung frame at time t inside phone k, with sung and spoken phone boundaries T_k and τ_k, is mapped back to the speech at

$$s(t) = \tau_k + (t - T_k)\,\frac{\tau_{k+1} - \tau_k}{T_{k+1} - T_k}, \tag{2}$$

and the HuBERT features are interpolated there. The melody comes in as a 256-bin pitch embedding of the gap-filled F0 on a mel-like scale plus a voicing embedding, and the speaker as a projected ECAPA embedding. These are added and passed through a Conformer (6 layers, $d = 256$, 4 heads, feed-forward 1024, kernel 31, dropout 0.1) and a linear layer with 128 mel outputs. The *pure* model predicts the mel directly. The *hybrid* model also gets the WORLD mel $\mathbf{M}_W$ and only predicts a correction,

$$\hat{\mathbf{M}} = \mathbf{M}_W + \mathbf{W}_o\mathbf{h} + \mathbf{b}_o. \tag{3}$$

Since $\mathbf{W}_o$ and $\mathbf{b}_o$ start at zero, training begins exactly at the WORLD output and the network only learns how real singing differs from it. Both models use an L1 loss (9.48M and 9.52M parameters). The pretrained OpenVPI NSF-HiFiGAN [7] turns the mel and the melody F0 into audio. Its copy-synthesis MCD on real singing is 1.0 dB, so the vocoder is not what limits quality. Whole songs are generated in 512-frame chunks with 128-frame cross-fades.

D. Other melodies and new recordings

To use another singer's version, we align the spoken phones with the sung ones by minimum edit distance (a greedy longest match paired repeated choruses with the wrong repeat). Matched phones copy their sung timing, an unmatched stretch shares its sung time among its spoken phones in proportion to their spoken durations, and one-sided stretches merge into a neighbour. Pairs with under 70% matched phones are refused (different songs never exceed 56%). Transposing by k semitones multiplies the F0 by $2^{k/12}$; another singer's melody is moved by whole octaves towards the speaker, $\Delta = 12\,\text{round}(\log_2(\bar{f}_{\text{spk}}/\bar{f}_{\text{mel}}))$, with $\bar{f}$ the median F0 of each person's speech. For new recordings, the Montreal Forced Aligner [12] (English ARPAbet models) derives both annotations from the lyrics, so a spoken take, a vocals-only sung take and the lyrics text are all the system needs.

E. Timing modes

Reference timing copies the annotated sung durations. *Predicted* timing stretches the predicted durations to fill the melody, but small errors add up over a song, and phrases end up starting 2.8 s away from where they should (median). The

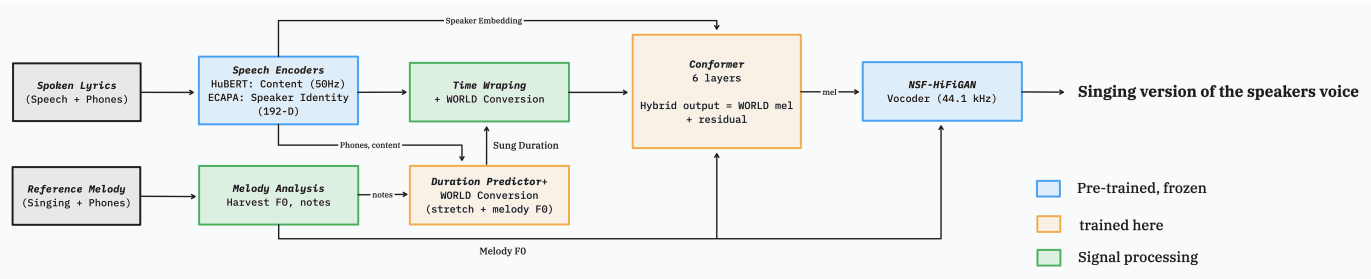


Fig. 2. System overview. Content and speaker come from the spoken lyrics, pitch and timing from the reference melody; the Conformer predicts the sung mel (in hybrid mode as a residual on a WORLD conversion's mel) and the NSF-HiFiGAN vocoder renders it with the melody F0.

phrase mode fixes this the way a score does: the melody's rests (pauses of at least 0.5 s) stay where they are, and the predicted durations are only rescaled within each phrase p of length T_p,

$$d'_i = \hat{d}_i \, T_p \Big/ \textstyle\sum_{j \in p} \hat{d}_j \qquad (i \in p). \tag{4}$$

This brings the median drift down to 0.33 s.

F. Post-filter

An L1 loss averages away detail, and in our hybrid it does so over time, not frequency: on the validation speaker its mel has 0.81 times the short-term temporal detail of real singing and its upper bands vary 0.8 times as much, while detail across frequency matches. An optional post-filter stretches each band around its song mean $\boldsymbol{\mu}$ by the ratio $\mathbf{g}$ of real to predicted standard deviations on that speaker (1.03 in the lowest to 1.35 in the upper bands, piecewise linear) and then applies a temporal unsharp mask,

$$\mathbf{M}' = \boldsymbol{\mu} + \mathbf{g} \odot (\hat{\mathbf{M}} - \boldsymbol{\mu}), \qquad \tilde{\mathbf{M}} = \mathbf{M}' + \alpha \, (\mathbf{M}' - \mathcal{S}_9 \mathbf{M}'), \tag{5}$$

with $\alpha = 0.25$ and $\mathcal{S}_9$ a 9-frame moving average.

IV. Experimental Setup

Training. We train the acoustic models with AdamW (learning rate $5 \cdot 10^{-4}$, one-cycle schedule) on batches of 32 random 512-frame crops in bf16, with gradient clipping at 1.0, on one RTX 6000 Ada, checking the validation L1 loss every 1000 steps and stopping after 8 checks without improvement. The pure model ran all 40,000 steps (23 min) but was best at step 9000; the hybrid stopped at 11,000 (6 min) and was best at 3000. The duration predictor ran 3000 steps on 128-phone windows (AdamW, 10^{-3}, batch 16).

Metrics. The mel-cepstral distortion (MCD) uses the coefficients c_1–c_{24} of an 80-band log-mel spectrum, averaged over the non-silent frames $\mathcal{F}$ of the real singing:

$$\mathrm{MCD} = \frac{10\sqrt{2}}{\ln 10} \, \frac{1}{|\mathcal{F}|} \sum_{t \in \mathcal{F}} \sqrt{\textstyle\sum_{k=1}^{24} \left(c_{t,k} - \hat{c}_{t,k}\right)^2}. \tag{6}$$

Frames are compared directly on the melody's timeline, so timing errors count, and this MFCC-style MCD is not comparable with other toolkits. F0 RMSE (in Hz, or in cents, $1200 \log_2(f/f_{\mathrm{ref}})$) and F0 correlation use Harvest on frames voiced in both signals. Speaker similarity is the ECAPA cosine between the output and the speaker's *spoken* recording. Real singing only reaches 0.44 (test) and 0.62 (validation) on it, so an output that still sounds like speech can score higher.

Validation gate. A variant replaces the hybrid only if, on the validation speaker, its MCD falls, its F0 RMSE rises by at most 0.5 Hz and its speaker similarity drops by at most 0.01; the test speaker is never used for decisions. Exp. 1 uses the own melody, Exp. 2 a different one, Exp. 3 unseen speakers (throughout), and Exp. 4 and 5 drop the melody or the speaker input.

V. Results

A. Duration prediction

On non-silence phones the duration error (validation / test, ms) is 174.5 / 156.5 for a global ratio (spoken $\times$ 2.30), 156.1 / 137.0 for a per-phone median ratio, 107.4 / 106.8 for the Transformer without notes and **92.8 / 90.8** with them: 34% better than the best rule on the test speaker, 15% of it from the notes. Most remaining error sits on the longest held notes, which the log-domain loss underestimates (Fig. 1, right).

B. Original melody and unseen speakers (Exp. 1 and 3)

Table I holds the main results. The *pure* Conformer has the lowest MCD but loses the speaker (0.00 on test) and tracks pitch worst (correlation 0.74): a new speaker's ECAPA embedding is an input it has never seen, so it falls back to an average voice. The *hybrid* gets the speaker back (0.35, against 0.44 for real singing), tracks pitch best (13.3 Hz, 0.93) and still beats WORLD on MCD; the order holds with predicted durations, though every MCD rises by 1.2–2.0 dB as the frame-aligned metric punishes timing errors. WORLD's high speaker score comes from keeping the *speech* timbre: in Fig. 3 it has sharp, buzzy harmonics all the way up, the pure model is blurred, and the hybrid keeps WORLD's structure while moving closer to real singing.

C. Ablations and attempts to improve the hybrid (Exp. 4 and 5)

Table II (F0 RMSE in Hz) shows what each input does. Without the melody input the pure model's F0 correlation falls from 0.74 to 0.41, while the hybrid hardly notices (0.93 to 0.92), as the WORLD mel already carries the melody's

TABLE I
OBJECTIVE RESULTS WITH THE SPEAKER'S OWN MELODY ON THE UNSEEN TEST AND VALIDATION SPEAKERS

	Test speaker (VKOW)				Validation speaker (JLEE)			
System	MCD (dB)↓	F0 RMSE (Hz)↓	F0 corr.↑	Speaker↑	MCD (dB)↓	F0 RMSE (Hz)↓	F0 corr.↑	Speaker↑
Reference durations								
WORLD baseline	4.32	17.6	0.891	**0.657**	4.01	17.8	0.916	**0.599**
Pure Conformer	**3.73**	29.2	0.739	0.000	**3.49**	26.9	0.756	0.069
Hybrid Conformer (ours)	4.12	**13.3**	**0.931**	0.354	3.64	**16.7**	**0.919**	0.401
Predicted durations (fully automatic)								
WORLD baseline	6.28	22.1	0.849	**0.645**	5.82	24.5	0.864	**0.630**
Pure Conformer	**5.04**	32.1	0.697	0.007	**4.72**	27.6	0.764	0.055
Hybrid Conformer (ours)	5.48	**17.7**	**0.891**	0.341	5.08	**21.8**	**0.883**	0.432
References								
NSF-HiFiGAN copy-synthesis	1.01	11.6	0.948	0.450	1.00	11.4	0.962	0.606
Real singing	0	0	1.000	0.436	0	0	1.000	0.616

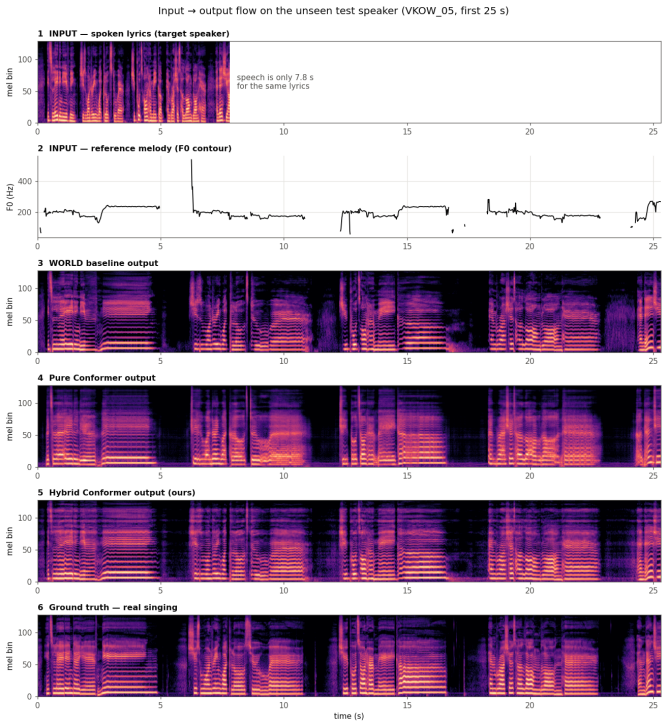


Fig. 3. The first 25 s of a test song: spoken input, melody F0, the outputs with reference durations, and the real singing.

harmonics. Dropping ECAPA *improves* the pure model's pitch (29.2 to 12.9 Hz), so the unseen speaker's embedding was confusing it; in the hybrid, ECAPA is what holds on to the voice, and without it speaker similarity halves (0.35 to 0.15) as the output slides towards a generic singing voice.

The hybrid sounds a little over-smoothed and overfits after about 3000 steps, so we tried four training fixes; none passed the gate. *Adversarial fine-tuning* (2-D patch discriminator on mel crops, least-squares GAN with feature matching, weights 0.1 and 0.5, 6000 steps) worsened every snapshot's MCD and raised speaker similarity by only 0.02–0.05, since sharper detail cannot be predicted frame by frame. An *EMA* of the weights (decay 0.999) only delayed the best checkpoint to

TABLE II
ABLATIONS AND HYBRID VARIANTS (REFERENCE DURATIONS)

	Validation			Test			
Model	MCD	F0	Spk.	MCD	F0	Corr.	Spk.
Pure, full	3.49	26.9	0.069	3.73	29.2	0.739	0.000
no melody	3.71	76.2	0.117	4.03	43.4	0.409	0.008
no speaker	3.37	12.2	0.186	3.73	12.9	0.936	0.110
Hybrid (final)	3.64	16.7	0.401	4.12	13.3	0.931	0.354
no melody	3.68	16.4	0.353	4.19	14.4	0.919	0.327
no speaker	3.33	11.7	0.206	3.68	11.3	0.952	0.153
GAN*	$\geq$4.07	$\geq$17.1	$\leq$0.45	–	–	–	–
EMA	3.66	16.1	0.270	4.43	13.0	0.934	0.296
EMA + noise 0.5	3.28	11.8	0.207	3.68	12.1	0.944	0.153
EMA + noise 1.0	3.30	13.3	0.204	3.70	12.9	0.935	0.150
lr 10^{-4}	3.81	17.2	0.564	4.21	16.9	0.893	0.615
post-filter†	3.75	17.0	0.465	4.21	15.3	0.912	0.426

*Over 6 snapshots: MCD 4.07–4.55, F0 17.1–20.3, Spk. 0.42–0.45. †Final hybrid + Eq. (5), run on CPU, where the unfiltered hybrid has F0 16.0 / 14.4 and corr. 0.919.

step 16,000. *Noise on the speaker embedding* (0.5 or 1.0 per-dimension standard deviations) landed next to the no-speaker ablation, as the model learned to ignore it. A *lower learning rate* stopped at step 1000, close to WORLD in timbre and MCD. All variants lie on one curve from "accurate spectrum, generic voice" to "speech-like voice, WORLD-like spectrum", with our hybrid in the middle (Fig. 4). The post-filter (Section III-F) lifts speaker similarity by 0.06–0.07 for 0.1 dB MCD but fails the gate too, so it stays an option at inference.

D. A different melody (Exp. 2)

Exp. 2 tests the real goal, an independently chosen melody. For each of the 8 validation and test songs we took every other singer's version (20 pairs; 14 moved down an octave), cut to the longest stretch where the annotations never differ by more than 5 phones at a time (166 s on average), and transposed the own melody by -3, 0 and $+3$ semitones. The hybrid follows the new melody best in every condition (Table III; the pure model stays at 272–364 cents and below 0.05 similarity): its pitch error stays within 4 cents of the own-melody value under a $\pm$3-semitone shift and within 25 cents with another singer's

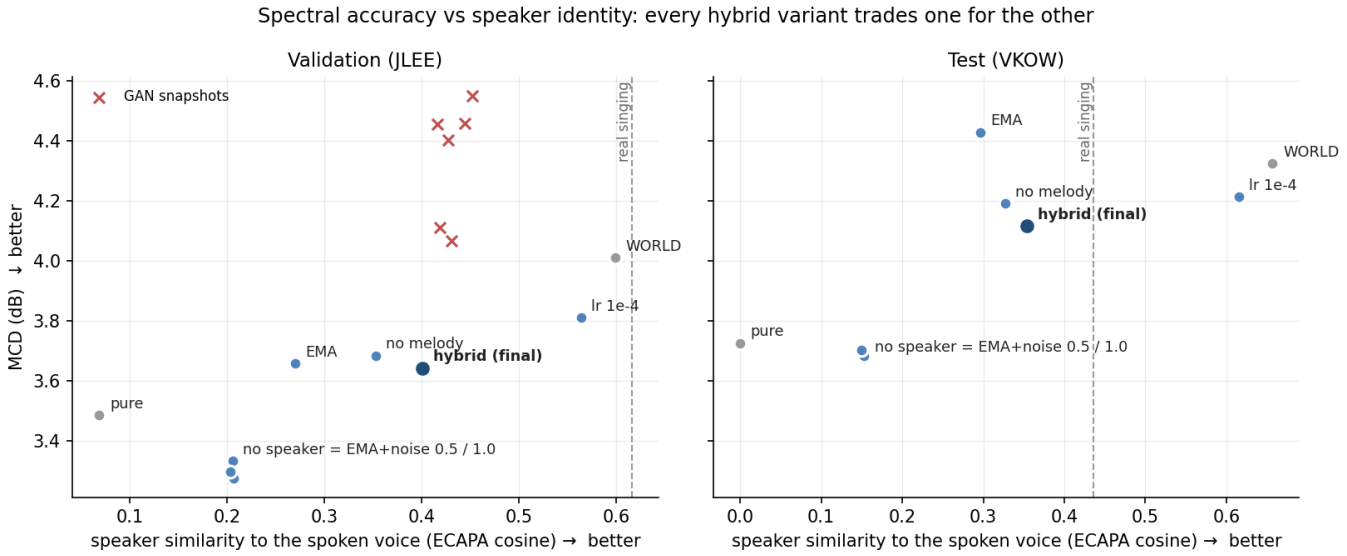


Fig. 4. MCD against speaker similarity for every system and training variant. A better variant would lie below and right of the final hybrid; none does.

TABLE III
EXP. 2, REFERENCE TIMING. LEAK.: SIMILARITY TO THE MELODY SINGER (OWN VOICES: 0.166)

Melody	System	F0 (cents)↓	Corr.↑	Spk.↑	Leak.↓
Own	WORLD	167	0.903	0.628	–
	Hybrid	**141**	**0.922**	0.377	–
Own −3 st	WORLD	153	0.903	0.681	–
	Hybrid	**144**	**0.909**	0.393	–
Own +3 st	WORLD	181	0.896	0.547	–
	Hybrid	**145**	**0.916**	0.342	–
Other singer (20 pairs)	WORLD	176	0.896	0.679	0.126
	Hybrid	**166**	**0.899**	0.408	**0.086**

TABLE IV
AUTHORS' OWN VOICES. SPK./OTHER: SIMILARITY TO OWN/OTHER'S SPEECH

Speaker	Key	Post-filter	Corr.↑	Cents↓	Spk.↑	Other↓
A (1st author)	singer's	–	**0.980**	**100**	0.428	**0.018**
	singer's	✓	0.965	141	**0.470**	0.107
B (3rd author)	singer's	–	**0.977**	122	0.293	0.364
	singer's	✓	0.960	**115**	0.382	0.357
	−12 st	–	0.953	132	0.303	0.275
	−12 st	✓	0.924	141	**0.391**	**0.257**

performance. The voice stays the speaker's: similarity matches the own melody (0.41 against 0.38), and leakage towards the melody singer (0.086) is below the 0.166 between the two singers' own voices in 18 of 20 pairs. Raising the pitch costs a little similarity (0.38 to 0.34 at +3); lowering it does not.

E. Fully automatic timing

Phrase timing is clearly the better automatic mode. Averaged over the 8 own-melody songs it lowers the MCD from 6.05 to 5.35 dB for WORLD, 4.88 to 4.46 for the pure model and 5.28 to 4.76 for the hybrid (reference timing: 4.17, 3.61, 3.88), closing about a third of the gap, and cuts the pitch error of WORLD (223 to 196 cents) and the hybrid (185 to 165; 203 to 175 with another singer's melody).

F. Our own voices

Two of us read the lyrics of an English song, with a 53-s a cappella take as the melody and reference timing; both were aligned automatically (95–96% of phones match) and nothing was tuned on them (Table IV). The output follows the melody and never resembles the singer (0.10 to A, 0.04 to B; A to B: 0.31). A scores 0.43–0.47, as high as real singing on the test speaker. B speaks much lower (median F0 112 against the singer's 229 Hz) and comes through clearly only an octave down with the post-filter, which raised every speaker score but cost up to 41 cents.

VI. DISCUSSION AND CONCLUSION

Correcting a WORLD conversion with a residual Conformer combines the strengths of both: WORLD supplies the speaker's timbre and the melody's harmonics, and the Conformer only learns how real singing differs. Every attempt to lower the MCD further moved the model towards an average voice, so ten training speakers seem to be the real bottleneck. The output still sounds somewhat muffled, with a little WORLD buzz that the post-filter only partly removes, and speaker similarity stays below real singing (0.35 against 0.44). Automatic timing costs 0.9 dB against reference timing, the melody must be a vocals-only take of the same lyrics, a key far from the speaker's range weakens the voice, and all results are objective single runs on four unseen speakers. Next we would run a listening test with at least ten listeners, pick the key from the speaker's range, and try pitch-shift and

tempo augmentation, more singers (e.g. NHSS [10]), a speaker reference encoder and onset-aware duration prediction.

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